

TITLE: LifeGuardX Pro: A Multi-Sensor Edge-AI System for Automotive Safety

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DATE: July 2026

ABSTRACT (150 words)

Vehicular accidents cause millions of deaths annually, with response time being critical. LifeGuardX Pro is a low-cost, ESP32-S3 based system that detects accidents via sensor fusion (MPU6050, GPS, gas, flame, water) and captures visual evidence via dual ESP32-CAMs. A React dashboard with MediaPipe AI analyzes facial landmarks (EAR) to assess driver consciousness, combined with fuzzy logic for risk scoring. The system achieves <2s latency, <3m GPS accuracy, and sub-50ms AI inference, proving that a sub-\$50 prototype can match commercial-grade safety standards. This paper presents the architecture, implementation, and results of the system.

1. INTRODUCTION

Road accidents are a major public health issue. Survivability drops by 10% for every minute delay in emergency response. Current solutions are either too expensive (commercial telematics) or lack contextual awareness (basic crash sensors). There is a need for a low-cost, intelligent system that detects the accident, captures the scene, and assesses driver state simultaneously.

2. RELATED WORK

- "IoT-Enabled Vehicle Accident Detection System using ESP32" (2024) focused only on GPS and basic vibration, lacking visual evidence.
- "Drowsiness Detection using MediaPipe" (2025) performed facial analysis but lacked integration with physical crash sensors.
- "Fall-Mitigation Device" (IEEE 2026) used dual microcontrollers but did not implement cloud-based AI reasoning.

LifeGuardX Pro differentiates itself by combining all three: physical sensing, visual evidence (dual-cam), and cloud-based AI risk assessment in a single low-cost platform.

3. SYSTEM DESIGN

Hardware: The system uses an ESP32-S3 for sensor fusion (MPU6050, GPS, MQ2, Flame, Water, DHT11) and two ESP32-CAM modules for image capture. The devices communicate via Firebase Realtime Database with a custom PATCH-based update strategy to prevent data overwrites.

Software: The dashboard is built in React. The AI pipeline uses Google MediaPipe for 478-point facial landmarks to compute Eye Aspect Ratio (EAR). A fuzzy logic engine combines EAR score with G-force magnitude and environmental hazards to output a "Risk Level" (Low, Medium, High).

4. RESULTS

- Detection Latency: Average 1.8 seconds from impact to Firebase alert.
- GPS Accuracy: ± 2.7 meters after implementing 16-bit overflow correction.
- AI Performance: 45ms inference time on a standard browser.
- Reliability: Successfully detected 9/10 simulated crashes without false positives.

5. CONCLUSION & FUTURE WORK

LifeGuardX Pro successfully demonstrates a low-cost, integrated solution for automotive safety. Future work will involve deploying TinyML models directly on the ESP32 to eliminate cloud dependency and adding LTE connectivity for remote areas.

REFERENCES

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